

Interview Video Analyzer - Code Lines Added and Analysis Flow

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Project: APITester (Angular)

1) Key Code Additions (Line References)

- Metric evidence model fields for timeline jump support
src/app/interview-verification/interview-verification.component.ts:9-10
src/app/interview-verification/interview-verification.component.ts:37-40
- Video reference and click-to-seek behavior
src/app/interview-verification/interview-verification.component.ts:51
src/app/interview-verification/interview-verification.component.ts:112-130
src/app/interview-verification/interview-verification.component.ts:132-138
- Core sampled interview frame analysis
src/app/interview-verification/interview-verification.component.ts:152-280
sample count logic: line 161; canvas scale: lines 164-165
- ROI extraction for mouth and both eyes
mouth ROI: lines 196-199
left eye ROI: line 203 onwards
right eye ROI: line 210 onwards
- Sustained eye-shift detection with blink filtering
src/app/interview-verification/interview-verification.component.ts:528-588
persistent motion validation: lines 590-611
- Audio window feature extraction (energy + zero crossing)
src/app/interview-verification/interview-verification.component.ts:283-399
window size 0.5s: line 304
silence threshold for pauses: line 330
- Risk formula mapping and weighted final score
src/app/interview-verification/interview-verification.component.ts:402-488
criterion risks: lines 403-409; weighted score starts line 464
- Tile-click UI wiring (HTML)
src/app/interview-verification/interview-verification.component.html:33
src/app/interview-verification/interview-verification.component.html:48-49
src/app/interview-verification/interview-verification.component.html:54
- Tile selected / hover / labels styling (CSS)
src/app/interview-verification/interview-verification.component.css:93-112
src/app/interview-verification/interview-verification.component.css:141-148
src/app/interview-verification/interview-verification.component.css:197

2) End-to-End Analysis Flow

- User uploads interview video and clicks Analyze Interview.
- System loads metadata and computes evenly spaced sample timestamps across full duration.
- For each timestamp, the video is seeked and a frame is rendered to 96x54 canvas.
- Frame difference is computed against previous sample frame to estimate motion.
- ROI differences are separately computed for mouth, left-eye region, and right-eye region.
- Scene switch events are counted when full-frame diff crosses high threshold.
- Eye shift events are detected using sustained high motion sequence logic with blink filtering.
- Audio track is decoded and processed in 0.5 second windows.
- Audio features: RMS energy, zero crossing rate, silence runs, overlap hints.
- Lip sync is measured by correlating normalized mouth-motion and aligned audio-energy series.
- Each criterion is converted to risk percent and clamped to 0-100.
- Overall risk is computed as weighted sum and classified into low-risk / needs-review / high-risk.
- For criteria with event timestamps, first occurrence time is attached to metric tile.
- When user clicks a tile, preview video seeks to that timestamp and starts playback.

3) Criterion-wise Analysis Logic

- Frequent eye shift away from camera: sustained motion in eye ROIs; short blink-like bursts filtered.
- Sudden change in answer style/depth: derived from audio energy variance.
- Long pauses before answer delivery: consecutive low-energy windows above threshold duration.

- Reading-like response pattern: cadence consistency from energy distribution stability.
- Background/overlapping voice hints: high energy + high zero crossing windows.
- Frequent screen/window switching: high full-frame diff spikes.
- Lip sync while answering: inverse of mouth-motion to audio-energy correlation.

4) Current Constraints / Notes

- This is sampled-timeline analysis, not every native frame at source FPS.
- Thresholds are heuristic and can be tuned per recording quality/environment.
- Face/eye ROIs are fixed geometric regions; no ML face landmarking is used yet.
- Best results require reasonably stable camera framing and clean microphone signal.

End of document